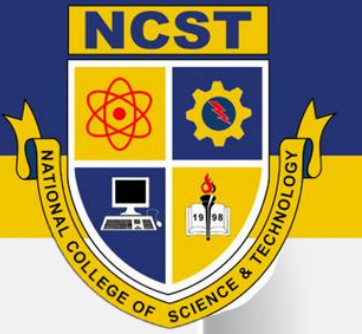




# **Ethical Principles in Information Security**

***ICT 310- Final Week 1***



# What is Ethics in Cybersecurity?

Ethics: **moral principles that guide behavior**

In cybersecurity, ethics ensures professionals **use knowledge and tools responsibly**

**Goal:** Protect data, systems, and people while avoiding harm

**Example:**

A security professional finds a vulnerability in a system. Reporting it responsibly instead of exploiting it is **ethical behavior**.



# Why Ethics Matter in Cybersecurity

Prevents harm to users and organizations

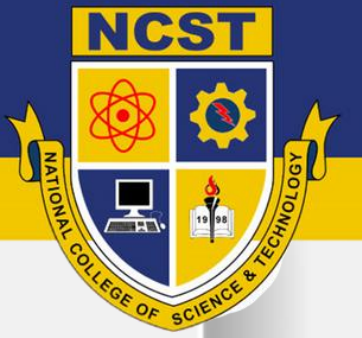
Builds **trust and credibility**

Guides decision-making in **complex situations**

Supports **legal and professional responsibilities**

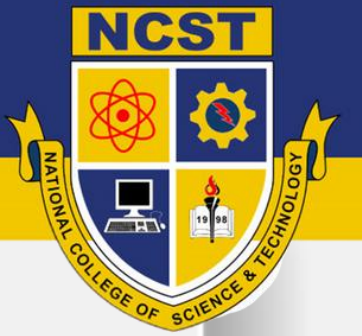
## **Example:**

A hacker could sell sensitive data but chooses to report it → ethical and professional action



# Core Ethical Principles

- 1. Confidentiality:** Respect private information
- 2. Integrity:** Maintain accuracy and honesty of data
- 3. Availability:** Ensure authorized access to data and systems
- 4. Accountability:** Take responsibility for actions
- 5. Professionalism:** Follow laws, standards, and best practices



# Common Ethical Dilemmas

An ethical dilemma is a situation where a person faces a **choice between two or more actions**, and **each option has moral or ethical consequences**

## Example:

You discover a **security flaw** in your school's grading system.

**Option 1 (Ethical):** Report it to the IT team so it can be fixed.

**Option 2 (Unethical):** Exploit it to change your grades.



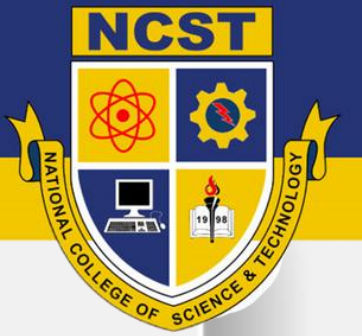


# Applying Ethics in Real-Life Cybersecurity Situations

- 1. Responsible Disclosure:** Reporting bugs or vulnerabilities instead of selling them.
- 2. Avoiding Unauthorized Access:** Never accessing systems or data without permission.
- 3. Data Protection:** Encrypting sensitive information to prevent leaks.
- 4. Fair Use of Tools:** Using cybersecurity tools for learning and protection, not for hacking.

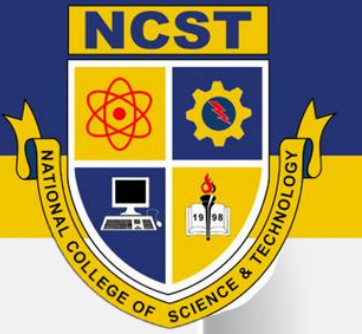


If you discovered a security flaw in your favorite app, what would you do and why?



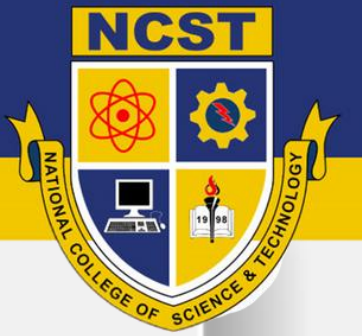
Ethics in cybersecurity is about **using your skills responsibly, protecting others, and making morally correct decisions, even when no one is watching.**





# **Professional Codes of Conduct for IT Professionals**

***ICT 310- Final Week 2***



# What Are Professional Codes of Conduct?

Guidelines that **define ethical behavior and responsibilities** for IT professionals.

Help professionals **make decisions that are legal, ethical, and socially responsible.**

Promote **trust, integrity, and professionalism** in the IT industry.

## **Example:**

Following a code ensures that software engineers do not release insecure apps.



# ACM Code of Ethics

**ACM (Association for Computing Machinery):** Global organization for computing professionals.

## **Key Principles:**

Contribute to society and human well-being

Avoid harm to others

Be honest and trustworthy

Respect privacy and confidentiality

Honor contracts, agreements, and professional responsibilities



# IEEE Code of Ethics

**IEEE (Institute of Electrical and Electronics Engineers):** Professional body for engineers and IT specialists.

## **Key Principles:**

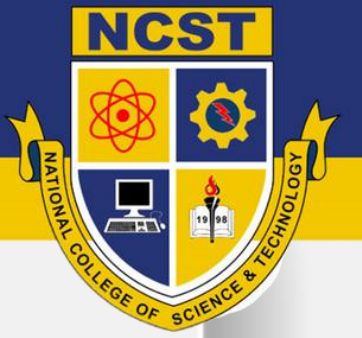
Hold safety, health, and welfare of the public as paramount

Avoid conflicts of interest

Be honest and realistic in stating claims

Treat colleagues and others fairly

Maintain and improve technical competence



# Responsibilities of IT Professionals

- Protect **confidential and sensitive data**
- Ensure **system reliability and security**
- Report **vulnerabilities and ethical issues responsibly**
- Maintain **professional competence and ongoing learning**
- Promote **ethical use of technology in society**

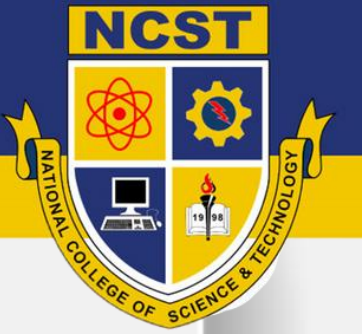
## Example:

A network administrator notices a colleague sharing passwords. Ethical responsibility: **report it to prevent harm.**



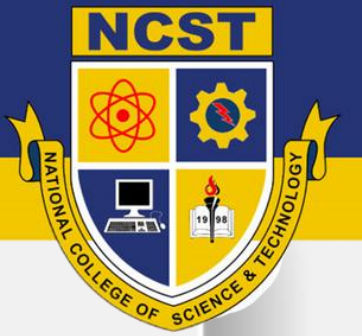


Why do you think following a professional code is as important as knowing technical skills?



# **Ethical Hacking and Responsible Disclosure**

***ICT 310- Final Week 3***



# What Is Ethical Hacking?

Ethical hacking is the legal and authorized practice of testing a computer system, network, or web application to find security vulnerabilities before malicious hackers (called black-hats) can exploit them.

Ethical hackers are also called “White-Hat Hackers” because they use their skills for defense, not attack.

They think like real hackers but act responsibly and with permission.

## **Example:**

A company hires a “penetration tester” to check if its website can be hacked.



# Purpose of Ethical Hacking

## **1. Identify Security Weaknesses**

Ethical hackers look for flaws in software, weak passwords, or open network ports that attackers could use.

## **2. Help Organizations Improve Security**

After finding vulnerabilities, they give detailed reports to help fix those weaknesses.

## **3. Prevent Cybercrimes**

Detecting issues early prevents data breaches, financial loss, and privacy violations.

## **4. Test Cyber Defenses**

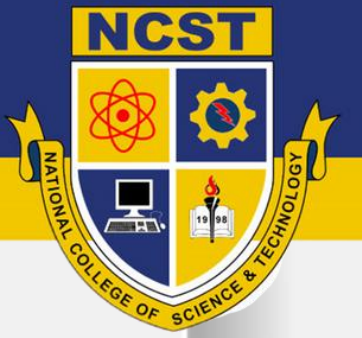
Ethical hackers simulate real-world attacks to see how well an organization can respond.



# Types of Hackers

| Type      | Description                                            | Goal                     |
|-----------|--------------------------------------------------------|--------------------------|
| White-Hat | Ethical hackers who follow rules                       | Improve security         |
| Black-Hat | Hackers who break laws                                 | Steal data or cause harm |
| Gray-Hat  | In between – may break rules but not for personal gain | Often seek recognition   |





# Difference Between White-Hat and Black-Hat Hacking

## White-Hat:

- ✓ Gets permission
- ✓ Follows laws
- ✓ Helps organizations fix issues

## Black-Hat:

- ✗ No permission
- ✗ Breaks laws
- ✗ Causes harm or steals data



# Why Ethical Hacking is Important

Identifies security holes before attackers do

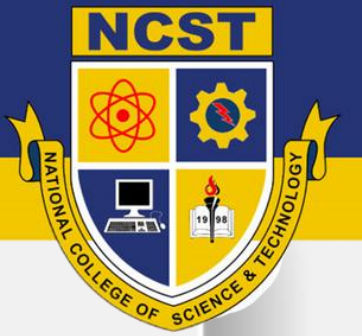
Protects sensitive data

Builds trust between users and organizations

Supports compliance with cybersecurity laws

## **Example:**

Government agencies often hire ethical hackers to test national defense networks.



# What is Responsible Disclosure?

Informing an organization privately about a security flaw before making it public, giving them time to fix it.

## **Steps:**

1. Discover the flaw
2. Report it responsibly
3. Wait for confirmation/fix
4. Publicly disclose (if allowed)

## **Example:**

A student finds a website bug and reports it to the admin instead of posting it online.



# Importance of Responsible Disclosure

Prevents exploitation by criminals

Encourages cooperation between hackers and companies

Promotes ethical behavior in the cybersecurity community

## **Example:**

Google's "Bug Bounty Program" rewards ethical hackers for responsible reports.



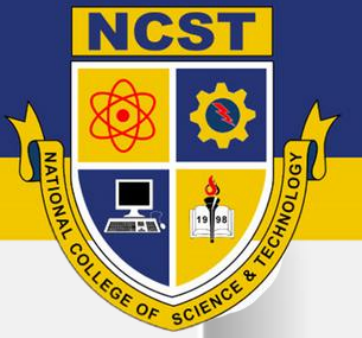
If you discovered a major bug in your favorite social media app, what's the most ethical action you can take?





# **Corporate Governance and Accountability**

***ICT 310- Final Week 4***



# What is Corporate Governance?

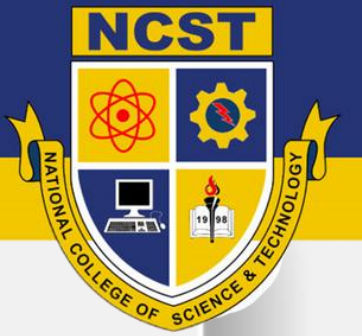
Corporate governance refers to the system of **rules, policies, and processes** that guide how an organization is managed and controlled.

## **In Information Security:**

It ensures that data protection, risk management, and ethical decision-making align with company goals.

## **Example:**

A company forms a **Cybersecurity Governance Board** to monitor data privacy and compliance.



# Importance of Corporate Governance

Builds trust among employees, clients, and the public

Promotes transparency and accountability

Ensures compliance with IT laws and standards (like ISO/IEC 27001, RA 10173)

Helps organizations respond properly to security breaches

## Example:

After a cyberattack, a company with clear governance policies quickly reports, investigates, and communicates the incident properly.



# What is Ethical Leadership?

Ethical leadership means leading by example — making fair, honest, and responsible decisions, even under pressure.

## **Key Traits:**

Integrity

Accountability

Transparency

Respect

## **Example:**

A team leader refuses to use unlicensed software, even if it saves money because it's unethical and illegal.



# Role of Ethical Leadership in IT

1. Sets a positive example for others to follow
2. Promotes a culture of honesty and responsibility
3. Ensures ethical use of technology and data
4. Encourages employees to report unethical behavior safely





# Organizational Policies

Policies are formal rules that describe how employees should behave and make decisions.

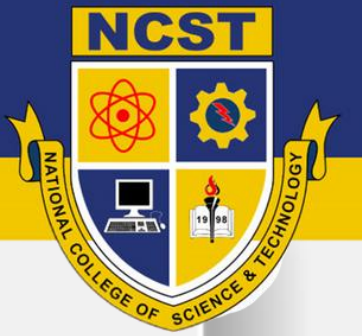
## **Examples of Common IT Policies:**

**Acceptable Use Policy:** Limits on company device usage

**Data Protection Policy:** Rules for handling personal data

**Incident Response Policy:** Steps to follow during a security breach

**Password Policy:** Minimum standards for strong passwords



# Reporting Ethical Violations

Reporting unethical actions or security policy breaches helps maintain accountability.

## Steps for Reporting:

**Identify** the unethical behavior or violation.

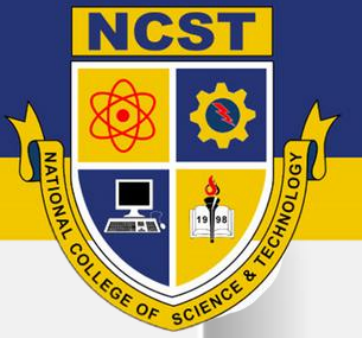
**Report** to your immediate supervisor or ethics officer.

**Document** evidence (emails, screenshots, logs).

**Maintain confidentiality** while following policy procedures.

## Example:

An employee reports a coworker who copied client data without permission.



# Handling Ethical Violations

## How Organizations Respond:

Conduct internal investigations

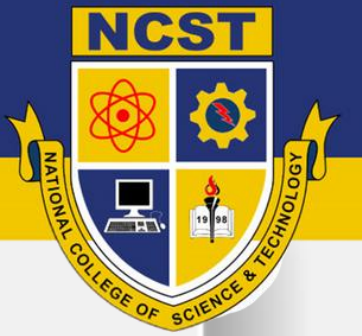
Take disciplinary actions (warnings, suspension, termination)

Provide ethics and compliance training

Review and update policies to prevent recurrence

## Example:

If an IT admin leaks customer data, HR and the IT Security team investigate and apply proper sanctions.



# Accountability in the Workplace

Accountability means taking responsibility for your actions and their outcomes — good or bad.

## **In IT Security:**

Every user is accountable for following data protection policies.

Admins must handle access rights carefully.

Managers are accountable for ensuring compliance.

## **Example:**

If a data breach happens, the IT head must explain the root cause and corrective measures.

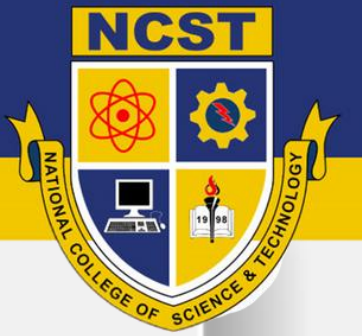


**If you discovered that a coworker was misusing customer data, how would you handle it ethically and professionally?**





# **Bonus Topic before ending the subject**



# Why We Need Information Assurance and Security (IAS)?

To help BSIT and BSCS students understand how to **protect data, systems, and users** in today's digital world.

## Why It Matters:

Builds a strong foundation in **cybersecurity**

Promotes **ethical and legal responsibility**

Prepares students for **real-world IT challenges**

Develops a **security mindset** — think like a hacker, act like a protector

## Example:

When managing networks or creating apps, you'll know how to secure data and prevent cyberattacks.



# The Goal of IAS for Future IT Professionals

IAS teaches you to be:

**Skilled** – able to detect, prevent, and respond to security threats

**Responsible** – follows IT laws and ethical standards

**Trusted** – protects users, companies, and communities online.